

MinJae Kim

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EDUCATION

- Sep. 2025 - **Stanford University**
PhD Student, Department of Materials Science and Engineering
(Advisor: Prof. Guosong Hong and Prof. Mark L. Brongersma)
- Feb. 2019 - **Korea Advanced Institute of Science and Technology (KAIST)**
Feb. 2025 *Bachelor of Science in Materials Science and Engineering*
- Total GPA of 4.22/4.3 (99.2/100), *Summa Cum Laude*, Honor student
 - Valedictorian and Representative Graduate at KAIST Commencement 2025 (Speech video)
 - Global Leadership Award (Top 18 out of ca. 11,000 students)
 - College of Engineering Leadership Award (Top 10 out of ca. 3,000 students)
 - Dean's List (Top 3% in College of Engineering)
 - Departmental Honor Scholarship (Top 4 out of ca. 120 students)
 - Fulfilled obligatory military service in Republic of Korea Army (Mar. 2021 - Sep. 2022)

PUBLICATIONS

2. M.J. Kim[†], J. Kim[†], S. Yoo^{*}, **Near-planar light outcoupling structures with finite lateral dimensions for ultra-efficient and optical crosstalk-free OLED displays**, *Nat. Commun.*, 2025, DOI: 10.1038/s41467-025-66538-6
1. M.J. Kim[†], D. Choi[†], C. Kang, S. Yoo^{*}, **Ultralow-power carbon dioxide sensor for real-time breath monitoring**, *Device*, 2025, DOI: 10.1016/j.device.2024.100681

HONORS AND AWARDS

Scholarships

- 2025 - 2030 **Kwanjeong Doctoral Study Abroad Fellowship**, Kwanjeong Foundation
- 2023 - 2024 **Woonhae Scholarship**, Woonhae Foundation
- 2023 **Young-Han Kim Global Leader Scholarship**, KAIST
- 2022 - 2027 **Dream Supporter Scholarship**, Global Hansang Dream Foundation
- 2021 - 2025 **KAIST Presidential Fellowship**, KAIST
- 2019 - 2025 **National Presidential Science Scholarship**, President of South Korea

Honors and Awards

- 2026 **First Place, Falling Walls Lab San Francisco Bay Area**, German Consulate General San Francisco
- 2026 **National Delegate to Global Young Scientists Summit**, National Research Foundation of Singapore
- 2025 **Valedictorian at KAIST Commencement 2025**, KAIST
- 2024 **National Delegate to 73rd Lindau Nobel Laureate Meeting**, Korean Academy of Science and Technology
- 2024 **NUS Young Fellow**, National University of Singapore
- 2023 **Young Future Energy Leader**, Khalifa University
- 2023 **Representative of KAIST, Young Engineers Honor Society**, National Academy of Engineering of Korea
- 2022 **Bronze Award**, Asian Students' Venture Forum
- 2021 **Talent Award of Korea**, Ministry of Education
- 2020 **Nobel Ceremony Guest and National Delegate**, Stockholm International Youth Science Seminar (SIYSS)
- 2020 **Cadet, Research Officer for National Defense**, Ministry of Defense, Ministry of Science and ICT

PATENTS

5. M.J. Kim, S. Yoo*, **Optical film comprising plurality of light-absorbing elements, and light-emitting diode comprising optical film** (KR-10-2025-0197889)
4. M.J. Kim, S. Yoo*, **Light emitting device comprising optical structure minimizing angular color shift and method for forming the structure thereof** (KR10-2025-0117210)
3. M.J. Kim, S. Yoo*, **Light emitting device for efficient light extraction and method for determining its structure** (KR10-2024-0201956)
2. M.J. Kim, H. Cho*, **Perovskite material including multi-functional asymmetric organic spacer and light-emitting device including the same** (KR10-2023-0118990)
1. M.J. Kim, S. Shin, H. Cho*, **Semiconductor material with passivation molecule giving π -conjugation electron density and method for manufacturing the same** (KR10-2023-0167999)

PRESENTATIONS

5. M.J. Kim, J. Kim, S. Yoo*, **Near-planar light outcoupling structure for ultra-efficient organic light-emitting diodes**, *UC Berkeley-Stanford Academic Conference* (Oral), 2025
🏆 *Best Presentation Award*
4. M.J. Kim, J. Kim, S. Yoo*, **Ultralow-power and highly stable carbon dioxide sensor for real-time breath monitoring**, *KAIST URP Workshop* (Oral), 2024
🏆 *Grand Prix*
3. M.J. Kim, J. Kim, S. Yoo*, **Near-planar light outcoupling structure for ultra-efficient organic light-emitting diodes**, *Optics and Photonics Congress* (Oral), 2024
🏆 *Best Paper Award*
2. M.J. Kim, S. Shin, H. Cho*, **Highly luminescent and stable quasi-2D perovskites based on multi-functional asymmetric spacer**, *KAIST URP Workshop* (Oral), 2023
🏆 *Grand Prix*
1. M.J. Kim, S. Shin, H. Cho*, **Highly luminescent and stable quasi-2D perovskites based on multi-functional asymmetric spacer**, *Spring Meeting of Korea Institute of Metals and Materials* (Poster), 2023
🏆 *Best Poster Presentation Award*

SKILLS

Language	English (fluent, TOEFL iBT: 106), Korean (native) L ^A T _E X(advanced), MATLAB (advanced), Python (moderate), HTML (moderate)
Simulation	LightTools (advanced), ORCA Density Functional Theory (intermediate), COMSOL Multiphysics (intermediate), RCWA (intermediate), Lumerical (novice)
Technical	Metasurface and plasmonics design and fabrication, Optical and photonic design of optoelectronics, PeLED/OLED fabrication and characterization, Organic synthesis and analysis, Confocal/fluorescence microscopy

REFERENCE

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